

Remark 1. *Convergence of the Markov process to the deterministic gradient flow describing trait dynamics occurs in the limit of small jumps $\sigma^2 \rightarrow 0$ [9]. In system (1), ε therefore acts as a small timescale separation parameter $0 < \varepsilon \ll 1$. On the slow timescale $\tau = \varepsilon t$ we recover the canonical equation of adaptive dynamics in the limit $\varepsilon \rightarrow 0$. Methods from geometric singular perturbation theory will be useful in the study of (1) for fixed $a \in A$ later.*

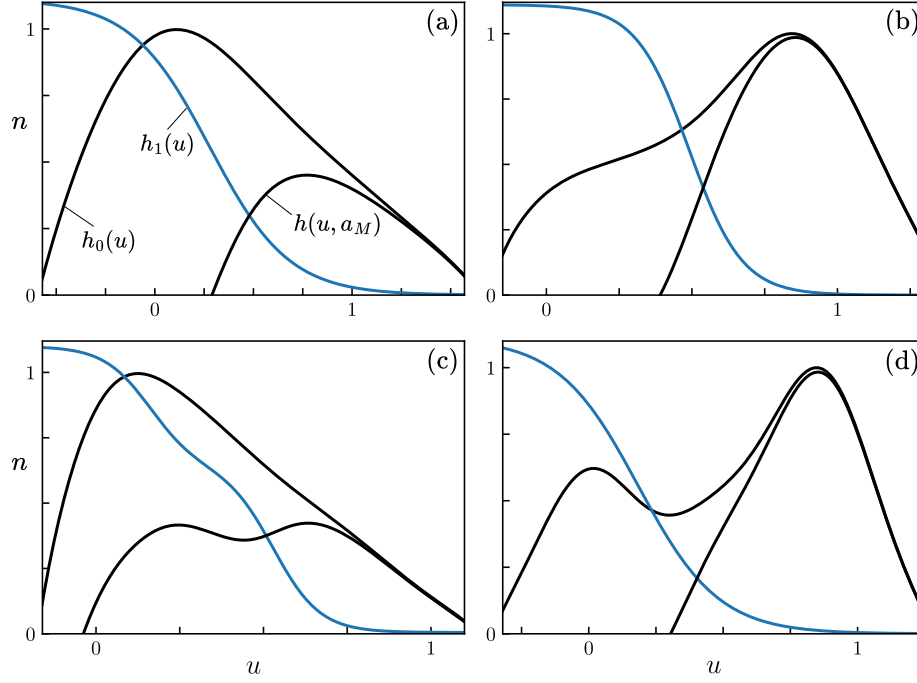


FIGURE 2. Examples of growth landscape $h(u, a) = h_0(u) - ah_1(u)$ (2) for treatment values $a \in \{0, a_M\}$. (a) Simple cost of resistance landscape. (b) An example without cost of resistance, where the growth landscape takes a unique maximum for high trait-values, remaining nearly unaffected by the treatment. (c) A more complex cost of resistance example. Under the effect of treatment, a single global maximum of the landscape is deformed into a “double well”-type potential at $a = a_M$. (d) A landscape with partial cost of resistance. It has a global maximum at high resistance for $a = 0$, shielded by a local maximum near $u = 0$. Parameter values and functions used can be found in Appendix B.

3. PRELIMINARY RESULTS

We write the system (1) more compactly as an affine control or switching system

$$(3) \quad \dot{x} = f(x, \alpha(t)) = f_0(x) + \alpha(t)f_1(x).$$